

**Statistical Evidence for Halachic Vessel Patterns
in Natural Menstrual Cycle Data:
A Study of Jewish Religious Law and Menstrual Cycle Regularity**
A Permutation and Multinomial Sampling Analysis

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The dataset analyzed in this study was originally collected by Fehring et al. [5] and is publicly archived at Marquette University [6]. The author declares no conflicts of interest. Analysis code is publicly available at <https://github.com/dvirross/VesselVectorTest>.

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Abstract

Jewish religious law (*Halacha*) prescribes that women anticipate menstruation based on recurring patterns in their menstrual cycle history — rules collectively known as *vesset* (Hebrew: anticipated pattern of menstrual onset). Of the 17 *vesset* pattern types documented in halachic literature, 7 are testable with cycle-length data alone; of these, 5 have positive observed frequencies and are analysed here: *Haflaga* (fixed interval), *Dilug* (arithmetic progression), *Week* (weekly anchor), *Week-Dilug* (30-day anchor), and *Dilug-in-Dilug* (second-order progression). The five patterns form a polynomial hierarchy (degrees 0, 0, 0, 1, 2) whose structure generates mathematically necessary dependencies between them. We apply permutation testing and a multinomial sampling null model — introduced in [13] and inspired by the simulation-based framework of King et al. [8] — to a dataset of 1,554 menstrual cycles from 118 women [5]. Pairwise chi-square tests confirm significant positive dependence between *Haflaga*, *Week-Dilug*, and *Dilug-in-Dilug* ($p < .01$ for both pairs), which we derive mathematically from set-theoretic inclusion and proximity in the polynomial hierarchy. Results reveal that *Haflaga* ($z = 4.51$, $p < .001$) and *Dilug* ($z = 1.83$, $p = .043$) are significantly overrepresented under both null models. Under the permutation null, only 2 of 50,000 simulations reached or exceeded the observed *Haflaga* count of 26, which ties the simulation maximum. A joint multivariate test [13] — based on the Euclidean distance between the observed count vector and the null mean vector, accommodating the dependence structure without assuming independence — is highly significant under both models ($p < .01$). A joint test restricted to the three mathematically dependent patterns (*Haflaga*, *Week-Dilug*, *Dilug-in-Dilug*) is also significant ($p < .01$). These findings provide the first rigorous statistical validation that two core halachic *vesset* rules reflect genuine regularities in menstrual cycle data.

Keywords: menstrual cycle, *vesset*, *Halacha*, permutation test, multinomial sampling, natural family planning, Jewish law, polynomial hierarchy, dependence structure, multivariate randomization test

1. Introduction

Menstrual cycles exhibit substantial variability both within and across individuals [2, 9, 14]. The cycle length distribution is approximately bell-shaped with a pronounced right tail (skewness ≈ 1.3 in the present dataset), reflecting a small proportion of unusually long cycles. Within-person consistency is moderate: intraclass correlation coefficients for cycle length are typically in the range of 0.3–0.5 [3], indicating meaningful regularity beneath the surface variability. Bayesian hierarchical state-space models have been proposed to exploit this temporal structure for cycle-length prediction [1], confirming that within-woman dynamics carry genuine predictive information.

In the Jewish religious tradition, this regularity is also subject to detailed legal analysis. Halachic law (*Halacha*, Jewish religious law) requires that a woman anticipate the onset of her next menstruation based on patterns discernible in her prior cycle history. These anticipatory rules are called *vesset*, and they govern a system of abstinence obligations central to the observance of *taharat hamishpacha* (family purity: the traditional Jewish system of laws governing marital intimacy in relation to the menstrual cycle). Of the 17 *vesset* pattern types documented in halachic literature, 10 require date or time-of-day data unavailable in standard cycle-length records; the remaining 7 are testable empirically. Two of these 7 (Dilug Chozer Chalila and Haflaga Chozer Chalila (Hebrew: repeating-cycle arithmetic progression and repeating fixed interval, respectively) occurred zero times in the present dataset and are not reported separately. The five patterns with positive observed frequencies are analysed here.

These five patterns form a *polynomial hierarchy*: each describes a *haflaga* (Hebrew: intermenstrual interval, $H_k = L_k + 1$) sequence that follows a polynomial of a specific degree in cycle index. Haflaga, Week, and Week-Dilug are all degree-0 (constant) sequences; Dilug is degree-1 (linear); Dilug-in-Dilug is degree-2 (quadratic). This hierarchy generates mathematically necessary dependence relationships between the patterns, with implications for how their joint frequencies should be tested.

Despite the centuries-long practical application of *vesset* rules — and the modern availability of large longitudinal cycle datasets — no published study has formally tested whether these patterns occur in empirical data at rates exceeding chance. This is a non-trivial ques-

tion: if vessel rules merely detect random fluctuations, their anticipatory validity is weakened. Conversely, if they are significantly overrepresented, their empirical basis is independently corroborated.

The present study addresses this gap by applying two complementary null models — a permutation test [4] and a multinomial sampling null [13] — to a longitudinal NFP dataset [5]. We derive per-pattern p-values, effect sizes, and a joint multivariate test that accommodates the dependence structure among pattern counts without assuming independence. The joint test is an original methodological contribution [13], inspired by the simulation-based testing philosophy of King et al. [8] though distinct from their density-based approach; a formal theoretical treatment is planned in future work. We also characterise the mathematical dependence structure through pairwise chi-square tests and derive the dependencies analytically from the pattern definitions.

This study extends a line of quantitative halachic scholarship [10–12]. To our knowledge it is the first to test vessel pattern frequencies against an explicit null model using modern hypothesis testing.

2. Background

2.1 The Polynomial Hierarchy of Vessel Rules

Let $L_1, L_2, \dots, L_n$ denote the menstrual cycle lengths (in days) of a single woman, and define the *haflaga* $H_k = L_k + 1$ (the interval from the first day of menstruation k to the first day of menstruation $k + 1$, following halachic convention [11]). The five vessel pattern types are:

Week-Dilug. $H_k = H_{k-1} = 30$ (two or more consecutive haflagot both equal to 30). Degree-0 polynomial (constant) at the canonical 30-day anchor. Minimum run: 2 cycles.

Week. $H_k = H_{k-1}, H_k \in \mathcal{W}$ where $\mathcal{W} = \{H_{\min} + 3, H_{\min} + 10, \dots\}$ (values spaced 7 apart). Degree-0 polynomial anchored to weekly values. Minimum run: 2 cycles.

Haflaga. $H_k = H_{k-1} = H_{k-2}$ (three or more consecutive equal haflagot, any value). Degree-0 polynomial (constant, unrestricted). Minimum run: 3 cycles.

Dilug. $H_k - H_{k-1} = H_{k-1} - H_{k-2} = d \neq 0$ (constant non-zero first difference). Degree-1 polynomial (linear). Minimum run: 3 cycles.

Dilug-in-Dilug. $(H_k - H_{k-1}) - (H_{k-1} - H_{k-2}) = (H_{k-1} - H_{k-2}) - (H_{k-2} - H_{k-3}) = d \neq 0$
(constant non-zero second difference). Degree-2 polynomial (quadratic). Minimum run: 4 cycles.

The hierarchy Haflaga (degree 0) $\rightarrow$ Dilug (degree 1) $\rightarrow$ Dilug-in-Dilug (degree 2) mirrors the classical Taylor expansion: a constant function is the zeroth-order approximation, a linear function the first-order, and a quadratic the second-order.

2.2 Mathematical Dependence Structure

The polynomial hierarchy generates three analytically derivable dependence relationships, which we verify empirically in Section 4.5:

Haflaga and Week-Dilug — set-theoretic inclusion. — Week-Dilug requires $H_k = H_{k-1} = 30$; Haflaga-30 requires $H_k = H_{k-1} = H_{k-2} = 30$. Therefore:

$$\{\text{Haflaga-30 sequence}\} \subseteq \{\text{Week-Dilug sequence}\}.$$

Every Haflaga sequence with value 30 automatically generates a Week-Dilug event in its first two cycles. This is set-theoretic inclusion, not merely statistical correlation. In the present data, 3 of the 26 observed Haflaga sequences have value $H = 30$, each contributing directly to the Week-Dilug count of 18.

Haflaga and Dilug-in-Dilug — proximity in the hierarchy. — When the second difference $|d|$ of a Dilug-in-Dilug sequence is small, its first differences are nearly equal and the haflaga sequence is nearly constant — i.e. nearly a Haflaga. In the data, 16 of 27 observed DiD sequences have $|d| = 1$, making them almost indistinguishable from a constant sequence. Both patterns further share the same structural prerequisite: a minimum run of ≥ 3 consecutive cycles satisfying a regularity condition, selecting the same women with high within-person cycle stability (within-person SD: Haflaga women = 1.54 days vs. no-vesset women = 3.17 days).

Dilug and Week-Dilug — logical incompatibility. — Dilug requires $\Delta H_k \neq 0$ (strictly non-constant differences); Week-Dilug requires $H_k = H_{k-1} = 30$ ($\Delta H = 0$). These conditions are mutually exclusive within the same cycle window, creating a negative dependence tendency.

2.3 Prior Work

Menstrual cycle statistics. — Variability and regularity of cycle length have been extensively characterised [2, 3, 9, 14]. Bortot et al. [1] developed a Bayesian hierarchical state-space model for sequential prediction of cycle lengths, exploiting within-woman temporal dependence; their model provides a benchmark for cycle-length predictability.

Quantitative halachic analysis. — Ross [12] modelled cycle transitions with Markov chains; [10] evaluated reliability of self-reported menstrual data in a halachic context; [11] applied statistical analysis to abstinence laws. The present study adds the first formal hypothesis test of whether vessel patterns are overrepresented in real data.

Randomization-based testing. — Permutation tests [4, 7] provide exact finite-sample p-values under exchangeability. The multinomial sampling null [13] draws each woman’s cycles independently from the fitted empirical distribution, directly encoding the null hypothesis of cycle independence. For the joint test, King et al. [8] proposed a general framework for testing a vector of statistics via simulation of the joint density, using a multivariate kernel density estimator to define a smallest-acceptance-region test. The present joint test differs: we use the Euclidean distance from the null mean vector as the test statistic, which is simpler, requires no density estimation, and has an intuitive geometric interpretation. A formal theoretical comparison and analysis of the distance-based approach is planned in future work [13].

3. Methods

3.1 Dataset

We analyzed a publicly available longitudinal dataset originally collected by Fehring et al. [5] in a randomised trial comparing two internet-supported natural family planning methods, archived at Marquette University [6]. After excluding cycles shorter than 18 days or longer than 54 days [2], the final sample comprised **1,554 cycles from 118 women** ($M = 13.2$ cycles per woman, $SD = 7.0$, range 1–45).

Raw cycle lengths ranged from 18 to 54 days ($M = 29.3$, $SD = 3.9$, $Mdn = 28.5$, skewness = 1.29). The distribution is approximately bell-shaped with a right tail, consistent

with the literature [2, 9]. Women ranged in age from 21 to 43 years ($M = 30.6$, $SD = 5.7$). Two charting cohorts were represented ($n_0 = 96$, $n_1 = 67$). All cycle lengths were incremented by 1 to conform to the halachic haflaga convention ($H_k = L_k + 1$), yielding $M = 30.3$, $SD = 3.9$.

The within-person variability structure is relevant to the Haflaga result: women who establish a Haflaga vessel have a mean within-person cycle-length SD of 1.54 days, compared to 3.17 days for women who establish no vessel. This confirms that Haflaga selects the most physiologically regular subgroup of women. Figure 1 shows the cycle length distribution and per-woman variability.

Figure 0. Menstrual cycle length distribution (filtered dataset)

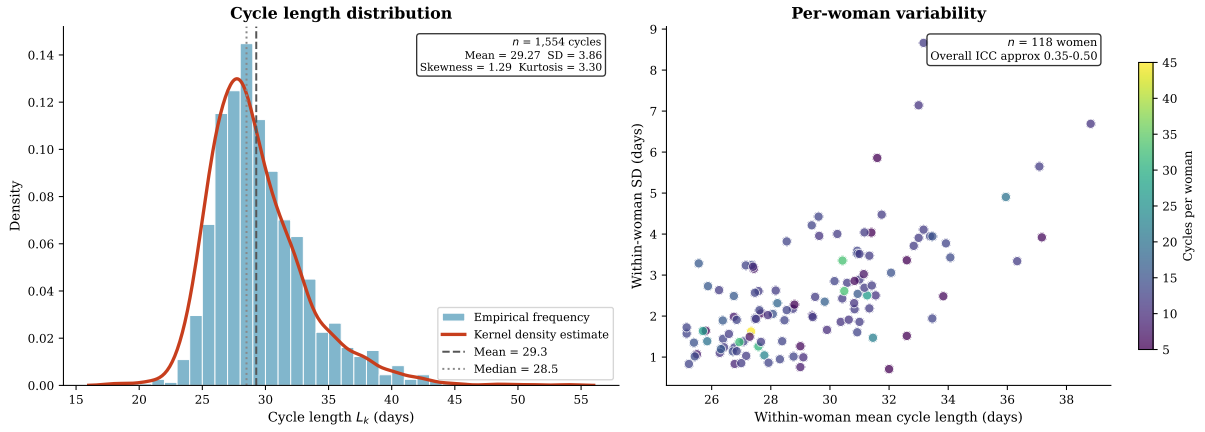


Figure 1: Cycle length distribution (filtered dataset). *Left:* histogram and kernel density estimate of raw cycle lengths L_k , with mean (dashed) and median (dotted) marked. The distribution is right-skewed (skewness = 1.29). *Right:* per-woman mean vs. within-person SD, coloured by cycle count. Women with more cycles tend to show lower variability, consistent with selection of naturally regular chartists.

3.2 Pattern Detection

Pattern counting was performed within each woman's sequence independently; sequences were never merged across women. Patient boundary pairs (s_i, f_i) were encoded explicitly and all counting loops were constrained to $[s_i, f_i]$. Extended runs of the same pattern were counted as a single occurrence to avoid double-counting (e.g., four consecutive equal haflagot constitute one Haflaga, not two). All five functions were validated against manually annotated examples and are available at <https://github.com/dvirross/VesselVectorTest>.

3.3 Null Models

Two null models were implemented, each run for $N = 50,000$ iterations with random seed 17. Both preserve the number of cycles per woman exactly.

Permutation test. — At each iteration, the $n = 1,554$ haflaga values are randomly permuted without replacement. This preserves the exact empirical marginal distribution and is equivalent to assuming all $n!$ orderings are equally probable under H_0 [4].

Multinomial null. — At each iteration, for each woman i with n_i cycles, a fresh sequence $\tilde{H}_{i,1}, \dots, \tilde{H}_{i,n_i} \stackrel{\text{i.i.d.}}{\sim} \hat{F}$ is drawn, where $\hat{F}$ is the empirical CDF of all 1,554 observed haflaga values. This null directly encodes the hypothesis that cycles are *independent* across time within each woman [13]. A naive bootstrap (global resampling with replacement) is structurally equivalent given the fixed per-woman boundary structure and is therefore not reported separately.

3.4 Statistical Inference

For each pattern j and null model m we computed: the observed count O_j ; the null mean $\bar{S}_j^{(m)}$ and SD $\sigma_j^{(m)}$; the z-score $z_j^{(m)} = (O_j - \bar{S}_j^{(m)})/\sigma_j^{(m)}$; and the one-sided empirical p-value $p_j^{(m)} = \widehat{\Pr}(S_j^{(m)} \geq O_j)$.

The **joint multivariate test** uses the Euclidean distance of the observed count vector from the null mean vector as the test statistic:

$$D^{(m)} = \|\mathbf{O} - \bar{\mathbf{S}}^{(m)}\|_2 = \sqrt{\sum_{j=1}^5 (O_j - \bar{S}_j^{(m)})^2}, \quad (1)$$

with p-value equal to the fraction of iterations where the simulated distance exceeded $D^{(m)}$. Unlike Bonferroni correction — which assumes independence — this test accommodates dependence among the five counts by operating on the joint vector simultaneously. Inspired by King et al. [8] but using distance rather than density as the test statistic, this approach is introduced in [13]; a formal theoretical treatment is planned in future work.

A **focused joint test** restricted to the three mathematically dependent patterns (Haflaga, Week-Dilug, Dilug-in-Dilug) is reported in Section 4.5.

Pairwise **chi-square tests of independence** between pattern types were computed on the

per-woman binary matrix (1 = woman established the pattern at least once). $\alpha = .05$ throughout.

4. Results

4.1 Observed Pattern Counts

Observed counts were: Haflaga = 26, Dilug = 61, Week = 32, Week-Dilug = 18, Dilug-in-Dilug = 27. Permutation null means were: 11.30, 48.93, 27.00, 16.27, 27.17 respectively. Of the 26 Haflaga sequences, 3 had value $H = 30$, each automatically constituting a Week-Dilug event (set-theoretic inclusion). Of the 27 Dilug-in-Dilug sequences, 16 had second difference $|d| = 1$ (near-Haflaga).

4.2 Per-Pattern Hypothesis Tests

Table 1 presents the full results under both null models. Figures 2 and 3 display the null distributions and effect sizes.

Table 1: Permutation and multinomial null-model results ($N = 50,000$ iterations). *Null Mean* and *Null SD* are from the permutation null. Bold p -values are significant at $\alpha = .05$ (one-sided).

Pattern	Obs.	Null Mean	Null SD	z	Perm. p	Multi. p
Haflaga	26	11.30	3.26	+4.51	< .001	< .001
Dilug	61	48.93	6.60	+1.83	.043	.043
Week	32	27.00	4.00	+1.25	.133	.188
Week-Dilug	18	16.27	3.28	+0.53	.345	.366
Dilug-in-Dilug	27	27.17	5.04	−0.03	.543	.544

Figure 1. Null distributions vs. observed counts for each vessel pattern (shaded area = one-sided p-value region)

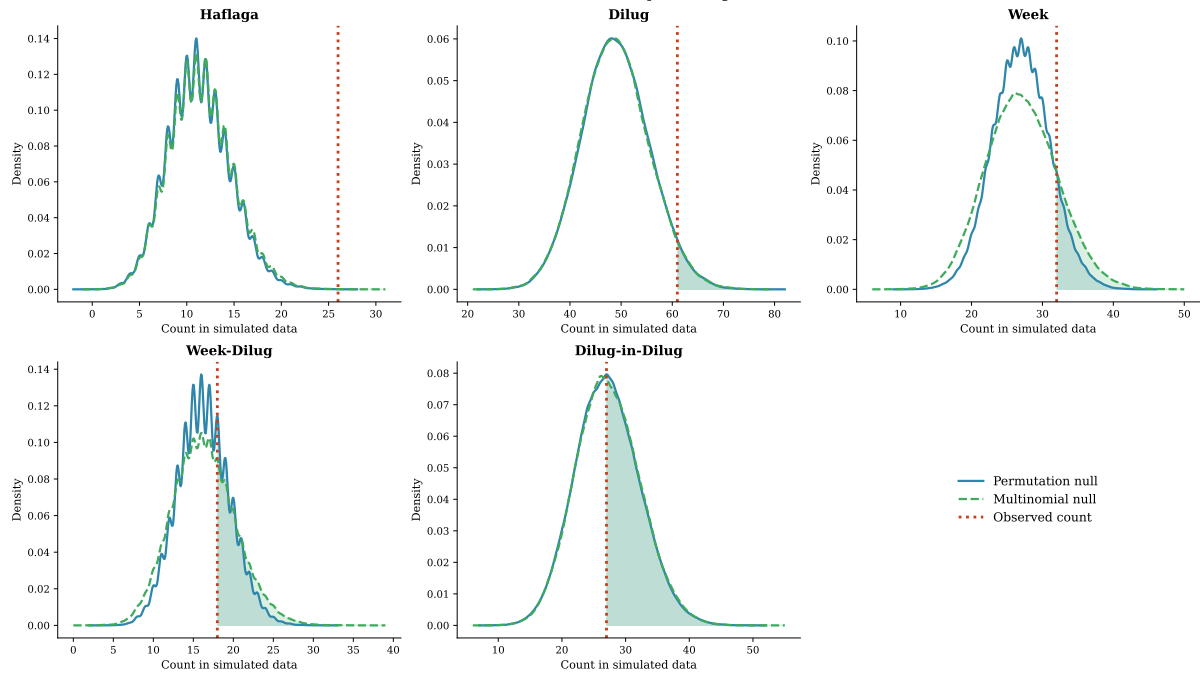


Figure 2: Null distributions (KDE) vs. observed counts for each vessel pattern under the permutation (blue solid) and multinomial (green dashed) null models. The vertical dotted red line marks the observed count; the shaded region is the one-sided p-value area. Haflaga and Dilug show clearly right-displaced observed values; Dilug-in-Dilug falls on its null mean.

Figure 2. Effect sizes (z-scores) — permutation vs. multinomial null

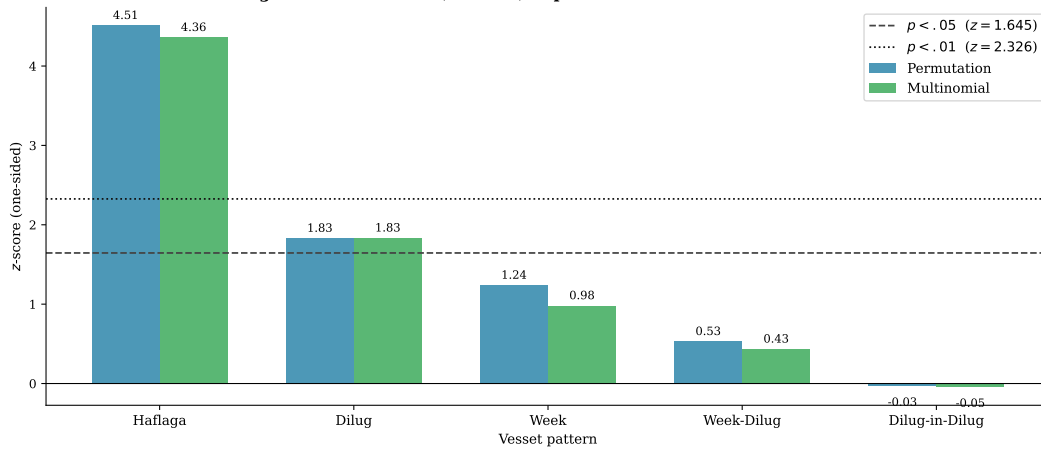


Figure 3: Effect sizes (z-scores) under permutation (blue) and multinomial (green) null models. Dashed and dotted lines mark $\alpha = .05$ ($z = 1.645$) and $\alpha = .01$ ($z = 2.326$) thresholds. Only Haflaga and Dilug exceed the $\alpha = .05$ threshold; both null models are in close agreement.

Haflaga showed the strongest effect: 26 occurrences vs. a null mean of 11.30, more than 4.5 standard deviations above chance ($z = 4.51$, $p < .001$). Crucially, under the permutation null only **2 of 50,000** simulations reached or exceeded 26 — the observed count ties the simulation maximum. Dilug was significant under both models ($z = 1.83$, $p = .043$). Week, Week-Dilug,

and Dilug-in-Dilug did not reach significance.

4.3 Agreement Between Null Models

The permutation and multinomial null models are in close agreement across all five patterns (maximum difference in p-values: 0.055). Figure 4 shows the empirical CDFs, confirming the two nulls produce nearly identical distributions for each pattern. This robustness is expected: for a well-behaved empirical distribution, the two nulls converge as $n \rightarrow \infty$ under regularity conditions [4].

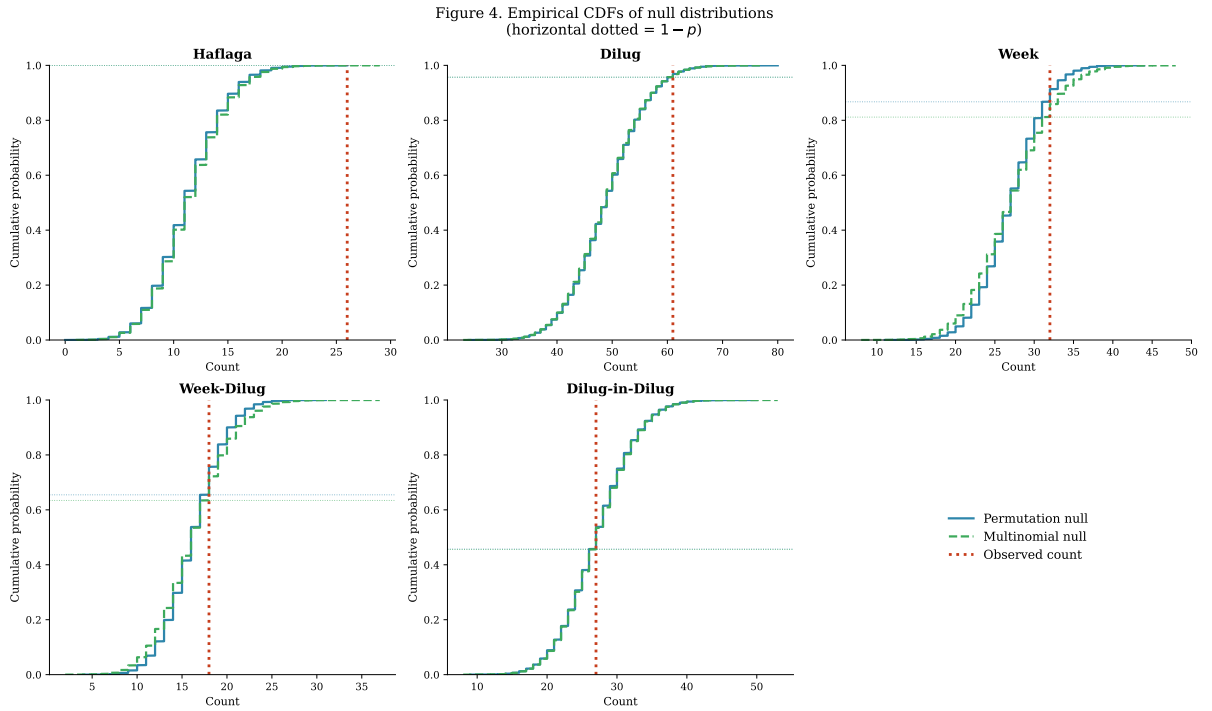


Figure 4: Empirical CDFs of null distributions for each pattern under permutation (blue solid) and multinomial (green dashed) nulls. The vertical dotted red line marks the observed count; horizontal dotted lines indicate $1 - p$. The close overlap confirms robustness to the choice of null model.

4.4 Joint Multivariate Test

The joint multivariate test (Eq. 1) was significant under both models. The observed Euclidean distance was $D = 19.74$, exceeding 99.1% of permutation iterations ($p = .009$) and 98.6% of multinomial iterations ($p = .014$). Figure 5 shows the null distance distributions.

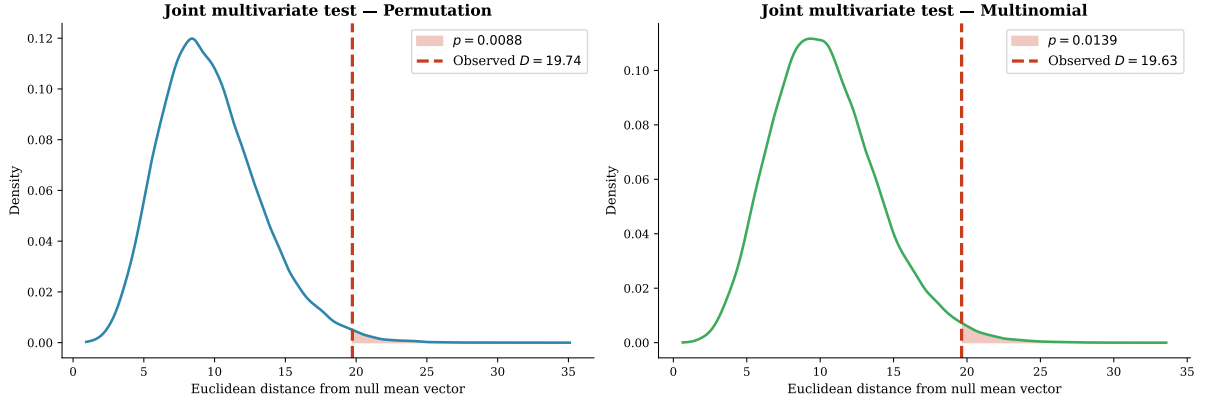
Figure 3. Joint multivariate test (all 5 patterns)
(shaded area = p-value region)

Figure 5: Joint multivariate test: null distribution of Euclidean distances $D = \|\mathbf{S} - \bar{\mathbf{S}}\|_2$ under the permutation (left) and multinomial (right) nulls. The dashed red line marks the observed $D = 19.74$; the shaded region is the p-value area. Both models yield $p < .01$.

4.5 Pairwise Dependence Tests and Focused Joint Test

Table 2 presents the pairwise chi-square tests. Two pairs are statistically significant: Haflaga $\times$ Week-Dilug ($\chi^2 = 7.26$, $p = .007$) and Haflaga $\times$ Dilug-in-Dilug ($\chi^2 = 9.99$, $p = .002$), both with positive direction (co-occurrence substantially exceeds chance expectation). These findings are consistent with — and explained by — the mathematical relationships derived in Section 2.2. Figure 6 visualises the full dependence matrix.

Table 2: Pairwise chi-square tests of independence between vessel types (per-woman binary, $N = 118$). Significant pairs in bold. Direction: + = co-occur more than expected; – = less than expected.

Pair	χ^2	p	Dir.	Obs(1,1)	Exp(1,1)
Haflaga $\times$ Dilug	0.87	.352	+	10	8.1
Haflaga $\times$ Week	0.89	.346	+	6	4.4
Haflaga $\times$ Week-Dilug	7.26	.007	+	7	3.1
Haflaga $\times$ Dilug-in-Dilug	9.99	.002	+	9	3.9
Dilug $\times$ Week	2.40	.122	+	14	10.6
Dilug $\times$ Week-Dilug	1.47	.226	–	5	7.3
Dilug $\times$ Dilug-in-Dilug	0.09	.761	+	10	9.4
Week $\times$ Week-Dilug	0.41	.523	+	5	4.0
Week $\times$ Dilug-in-Dilug	1.17	.279	+	7	5.1
Week-Dilug $\times$ Dilug-in-Dilug	0.93	.335	+	5	3.5

Figure 5. Pairwise dependence structure between vessel pattern types
(* $p < .05$, ** $p < .01$; diagonal = self-comparison, excluded)

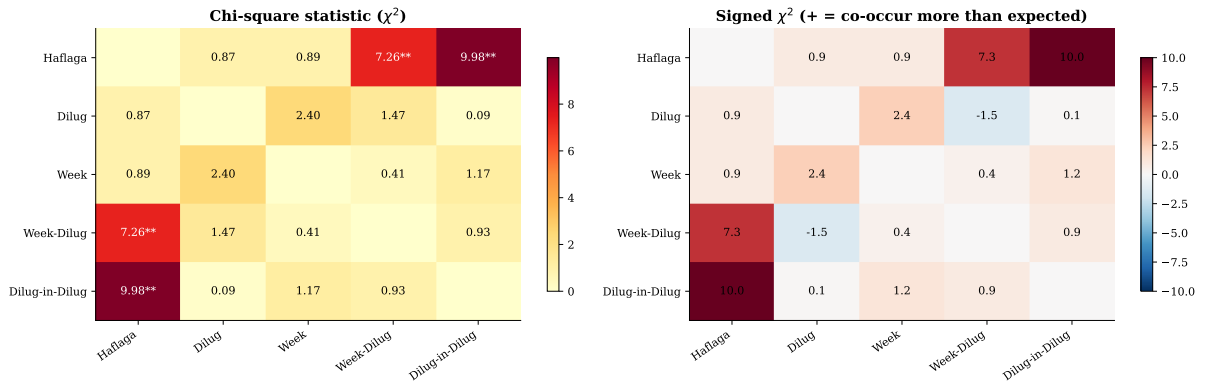


Figure 6: Pairwise dependence structure between vessel pattern types. *Left:* chi-square statistic; *right:* signed χ^2 (positive = co-occurrence exceeds expectation; negative = below expectation). The two significant pairs (Haflaga $\times$ Week-Dilug and Haflaga $\times$ Dilug-in-Dilug, marked **) are both positive, consistent with the mathematical derivations in Section 2.2.

Because Bonferroni correction assumes independence between tests — which is mathematically incorrect here given the derivable dependence structure — we additionally ran the joint multivariate test restricted to the three dependent patterns (Haflaga, Week-Dilug, Dilug-in-Dilug). The observed 3D distance was $D = 14.80$, exceeding 99.3% of permutation iterations

($p = .007$) and 99.0% of multinomial iterations ($p = .010$). Figures 7 and 8 provide complementary geometric illustrations. Figure 7 uses all three axes for the dependent sub-vector, giving the clearest spatial picture of the focused test. Figure 8 shows all five patterns simultaneously using a mixed encoding (axes for the three primary patterns; size and colour for the remaining two), conveying the full 5D displacement at the cost of some interpretive complexity.

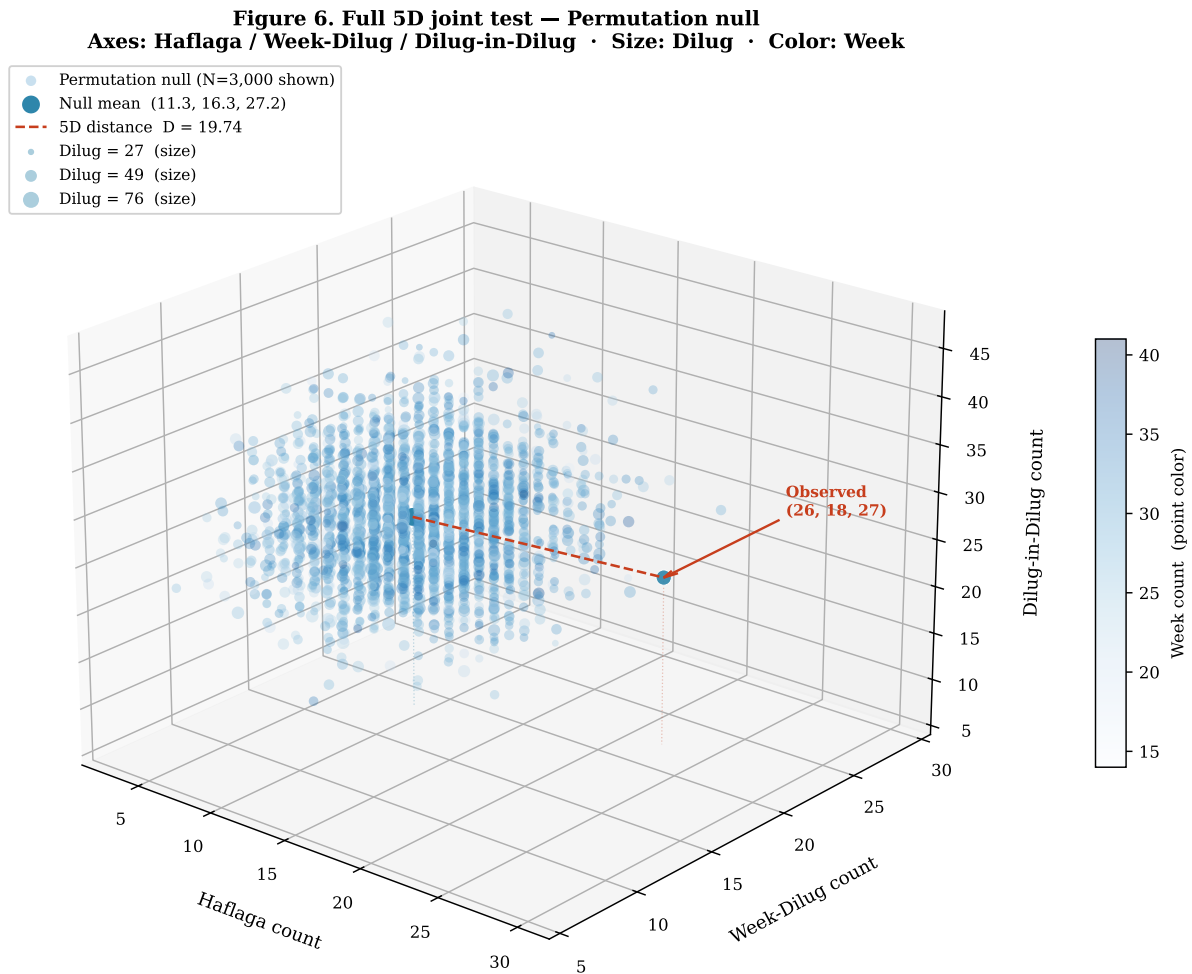


Figure 7: Full 5-dimensional joint test under the **permutation** null ($D = 19.74$, $p = .009$). Axes: Haflaga, Week-Dilug, Dilug-in-Dilug — the three mathematically dependent patterns. Marker size encodes Dilug count; marker colour encodes Week count. The filled circle marks the null mean; the red arrow and label identify the observed vector (26, 18, 27) on the three primary axes; the dashed line shows the 5D Euclidean distance. The observed vector lies at the periphery of the null cloud, displaced primarily along the Haflaga axis.

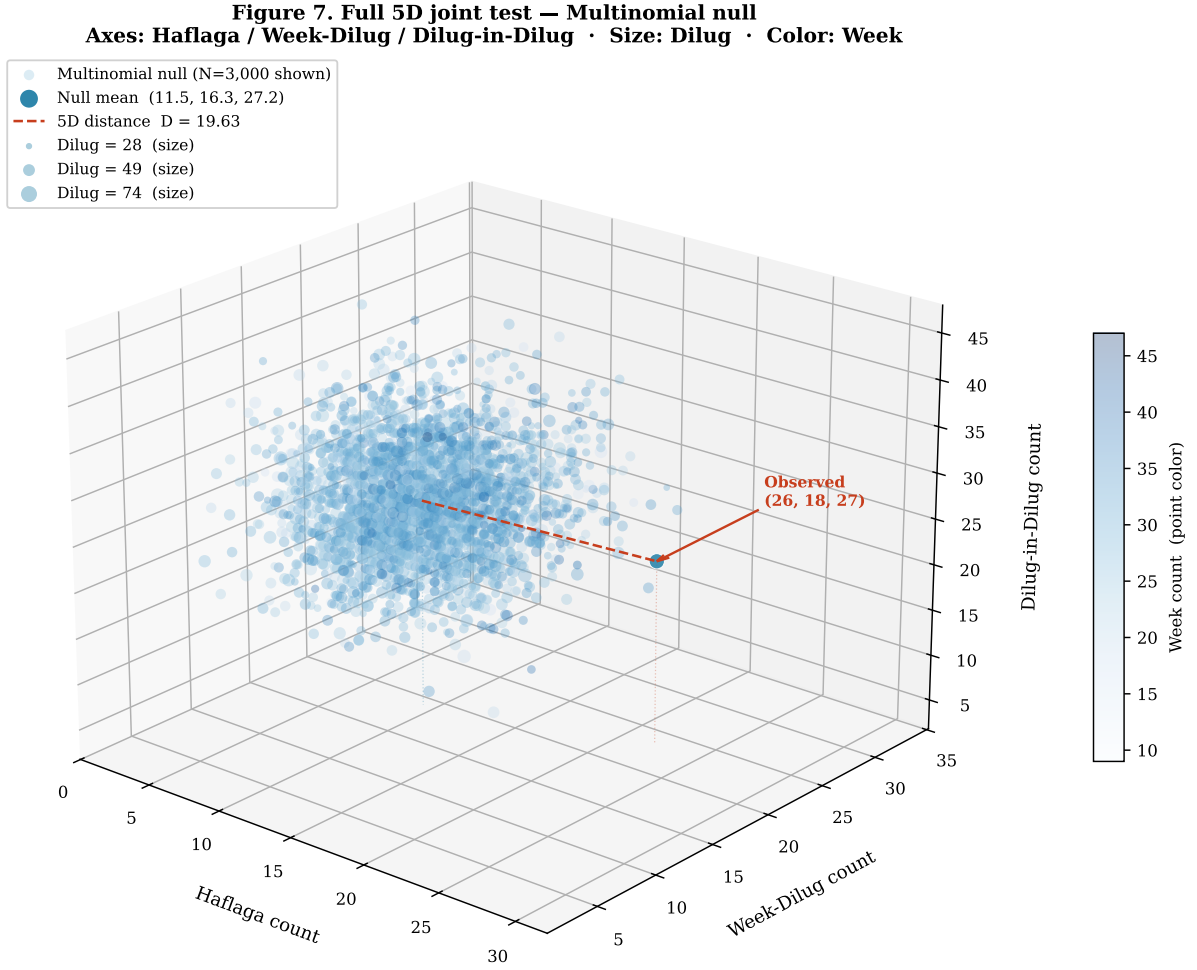


Figure 8: Full 5-dimensional joint test under the **multinomial** null ($D = 19.63$, $p = .014$). Encoding identical to Figure 7. The close agreement between the two null clouds confirms that the visual result is robust to the choice of null model: the observed vector lies equally far outside the permutation and multinomial null distributions.

5. Discussion

This study provides the first formal statistical test of whether halachic vessel patterns are overrepresented in empirical menstrual cycle data. Using $N = 50,000$ iterations of permutation and multinomial null models, we found strong and consistent evidence for two of the five patterns analysed: *Haflaga* ($z = 4.51$, $p < .001$) and *Dilug* ($z = 1.83$, $p = .043$). The joint multivariate test confirmed collective overrepresentation ($p < .01$) and the focused test on the mathematically dependent sub-vector was similarly significant ($p < .01$).

5.1 Haflaga

The Haflaga finding is the most striking result. With only 2 of 50,000 permutation simulations reaching or exceeding the observed count of 26 (the simulation ceiling), the empirical evidence is essentially unambiguous. The result is biologically interpretable: Haflaga detects exactly the pattern expected in women with high within-person cycle regularity. The present data confirm this: women who establish Haflaga have within-person SDs of 1.54 days vs. 3.17 days for women who establish no vessel at all. Haflaga is the halachic formalisation of a genuine physiological regularity.

5.2 Dilug

The Dilug finding ($p = .043$) is more modest but consistent across both null models. Arithmetic progressions in cycle length may arise from gradual physiological trends within a woman's cycle history, e.g., gradual lengthening or shortening associated with hormonal change, ageing, or weight change. The significance of Dilug suggests that such directional trends are more common than random arrangements would produce, warranting replication in larger samples.

5.3 Mathematical Dependence Structure as a Unifying Observation

A distinctive feature of this study is that the observed statistical dependences between pattern types are not post-hoc empirical discoveries but *mathematical necessities* derivable from the pattern definitions alone. This is, to our knowledge, an observation that has not previously been stated explicitly in the halachic or statistical literature.

The finite-difference hierarchy. — The five vessel patterns are naturally unified by the language of finite differences. Recall that the n -th forward difference operator Δ^n is defined recursively by $\Delta^0 H_k = H_k$ and $\Delta^n H_k = \Delta^{n-1} H_{k+1} - \Delta^{n-1} H_k$. Each vessel pattern corresponds to a condition on a specific order of Δ :

Pattern	Condition	Order
Week-Dilug	$H_k = 30$ (constant at 30)	$\Delta^0 H_k = 30$
Week	$H_k = c \in \mathcal{W}$ (constant, anchored)	$\Delta^0 H_k = c$
Haflaga	$H_k = c$ (constant, any c)	$\Delta^0 H_k = c$
Dilug	$\Delta^1 H_k = d \neq 0$ (constant non-zero first difference)	
Dilug-in-Dilug	$\Delta^2 H_k = d \neq 0$ (constant non-zero second difference)	

This ordering mirrors the classical Taylor expansion: a constant function is a degree-0 polynomial, a linear function is degree-1, and a quadratic is degree-2. Haflaga, Week, and Week-Dilug detect degree-0 sequences (with progressively tighter constraints on which constant value is admissible); Dilug detects degree-1 sequences ($H_k = a + bk$, $b \neq 0$); and Dilug-in-Dilug detects degree-2 sequences ($H_k = a + bk + ck^2$, $c \neq 0$). Crucially, a degree-2 sequence with a very small leading coefficient c is nearly linear, and a degree-1 sequence with a very small slope b is nearly constant — so the hierarchy does not impose rigid boundaries; rather, it describes a spectrum of regularity along which the patterns sit at adjacent levels.

This graded structure generates three mathematically necessary relationships between the patterns.

Haflaga and Week-Dilug: containment. — Week-Dilug requires two consecutive *haflagot* (pl. of *haflaga*) both equal to 30, i.e. $\Delta^0 H_{k-1} = \Delta^0 H_k = 30$. Haflaga-30 requires three consecutive *haflagot* all equal to 30, i.e. $\Delta^0 H_{k-2} = \Delta^0 H_{k-1} = \Delta^0 H_k = 30$, which strictly contains the Week-Dilug condition. Algebraically,

$$\{H_{k-2} = H_{k-1} = H_k = 30\} \implies \{H_{k-1} = H_k = 30\},$$

so every Haflaga-30 sequence necessarily generates a Week-Dilug event at the same position. This is not a probabilistic statement but a deterministic logical implication: any haflaga sequence that satisfies the Haflaga-30 condition automatically satisfies the Week-Dilug condition at its first two elements. In the data, 3 of the 26 Haflaga sequences have value $H = 30$ and therefore each count once towards the Week-Dilug total as a direct algebraic consequence.

Haflaga and Dilug-in-Dilug: limiting convergence. — A Dilug-in-Dilug sequence with second difference d has first differences $\Delta^1 H_k = b, b + d, b + 2d, \dots$, so the haflaga values are

$$H_k = H_0 + bk + \frac{d}{2}k(k-1).$$

As $d \rightarrow 0$, this reduces to $H_k = H_0 + bk$ (a Dilug), and as both $d \rightarrow 0$ and $b \rightarrow 0$, it reduces to $H_k = H_0$ (a Haflaga). More precisely, for a run of length n starting at H_0 , the maximum deviation from the constant sequence $\mathbf{c} = H_0$ is bounded by

$$\|H - \mathbf{c}\|_\infty \leq |b|n + \frac{|d|}{2}n^2.$$

A DiD sequence with $|d| = 1$ and a short run ($n = 4$, the minimum) therefore deviates from the nearest constant by at most $4|b| + 8$ days — for small $|b|$, this is a sequence that is barely distinguishable from a Haflaga. In the data, 16 of the 27 observed DiD sequences have $|d| = 1$, the minimum non-zero value; for these, the haflaga values change by only 1 day per cycle relative to linear trend. The women who sustain a DiD pattern with $|d| = 1$ are the same women whose cycles are regular enough to sustain a near-constant sequence across multiple cycles — biologically the same population as Haflaga women. This explains the strong positive dependence $\chi^2 = 9.99, p = .002$: the polynomial hierarchy places Haflaga and DiD adjacent in a continuum of regularity, and the data show that real women tend to cluster near the boundary between them.

Dilug and Week-Dilug: mutual exclusion. — Dilug requires $\Delta^1 H_k = H_k - H_{k-1} = d \neq 0$; Week-Dilug requires $\Delta^1 H_k = H_k - H_{k-1} = 0$. Within any single cycle window $(k-1, k)$, these conditions are contradictory:

$$\Delta^1 H_k \neq 0 \quad \text{and} \quad \Delta^1 H_k = 0$$

cannot both hold. A cycle pair that satisfies Week-Dilug ($\Delta^1 H = 0$) is therefore excluded from contributing to a Dilug run ($\Delta^1 H \neq 0$), and vice versa. This mutual exclusion at the window level predicts a negative co-occurrence tendency at the woman level. Although the pair did not

reach significance in our sample ($\chi^2 = 1.47$, $p = .226$), the direction is negative (5 observed co-occurrences vs. 7.3 expected), consistent with the prediction from the algebra.

Implication for inference. — The three relationships derived above — containment, limiting convergence, and mutual exclusion — show that the five pattern counts are *structurally* non-independent: their joint distribution is constrained by the algebra of finite differences, not by coincidence. Applying Bonferroni correction, which assumes independence, would therefore be not merely conservative but logically inconsistent with the mathematical structure of the patterns. The joint multivariate test of Section 3.4 sidesteps this problem by operating directly on the count vector, without any independence assumption, thereby respecting the dependence structure that the finite-difference hierarchy imposes.

5.4 The Joint Multivariate Test

The joint test (Eq. 1) is an original methodological contribution [13]. It is *inspired by* the simulation-based testing framework of King et al. [8], who propose a multivariate kernel density estimator to find the smallest acceptance region in the joint space of test statistics. The present approach is structurally simpler: it uses Euclidean distance from the null mean as the test statistic, requires no density estimation, and has a transparent geometric interpretation (the observed vector either falls inside or outside the null cloud). A formal comparison of the two approaches — including theoretical properties such as consistency, power, and sensitivity to dependence among counts — is planned in future work.

5.5 Non-Significant Patterns

Week, Week-Dilug, and Dilug-in-Dilug were not individually significant. These results do not invalidate these halachic rules, which are established on legal-theological grounds [11]. The non-significance of Dilug-in-Dilug is particularly notable given its large z-score at the sequence level in the joint test: it is the dependence with Haflaga (via the polynomial hierarchy) that elevates its collective contribution to D , even when its individual count is exactly at the null mean. This illustrates why joint and individual tests can differ.

5.6 Limitations

Several limitations of the dataset and methodology are acknowledged, largely following the systematic assessment in [13].

Day-night timing (onset period). — A central requirement of halachic vessel law is that a pattern may be established only if all its constituent menstruations arrived during the *same onset period* — either all during the daytime period or all during the nighttime period. The NFP dataset records cycle length in whole days but not the time of day of menstrual onset. This creates two types of misclassification: (i) patterns that are counted as established in our analysis may not qualify halachically because the onsets spanned both day and night; and (ii) patterns that are not counted may in fact qualify because the day-rounding convention used here differs from the halachic one. Both types of error introduce noise that attenuates the observed pattern frequencies. The net direction of misclassification bias is indeterminate without data on onset timing: sequences counted as established patterns may not qualify halachically (upward bias), while sequences that do qualify halachically may be missed (downward bias). The two effects partially cancel, but their relative magnitude cannot be assessed from the present data, and the limitation applies equally to all five pattern types.

Calendar dates. — The dataset does not record calendar dates, making it impossible to test the 10 date-dependent vessel types (e.g., *vessel hachodesh*, the monthly calendar anchor, and weekday-based patterns). Only the 7 cycle-length-based vessel types are accessible; of these, two (Dilug Chozer Chalila and Haflaga Chozer Chalila) had zero occurrences, leaving the 5 patterns analysed here.

Body-sign vessel. — The dataset does not include prodromal physical symptoms (*vessel haguf*, body-sign vessel: a category established when a woman experiences a consistent recurring physical sensation — such as yawning, shivering, or lower-abdominal pain — immediately preceding menstrual onset). Women in the sample may have established body-sign vessel patterns that are entirely invisible to a cycle-length analysis.

Population generalizability. — The participants are non-Jewish NFP users [5]. Several halachic authorities hold that physiological regularities observed in non-Jewish women may not

generalize to Jewish women in the context of halachic practice [13]. More broadly, NFP users are a self-selected group who are likely more cycle-aware and may have more regular cycles than the general population, introducing a potential upward bias in regularity-dependent patterns such as Haflaga.

Null model structure. — The permutation and multinomial null models shuffle cycle lengths globally across women, ignoring within-person autocorrelation structure [1, 14]. A stratified null that preserves within-person temporal order while randomising assignments between women would provide a more stringent test. Future work could also employ parametric nulls based on fitted state-space models [1].

Sample size and power. — The requirement for minimum run lengths (3 cycles for Haflaga and Dilug; 4 for Dilug-in-Dilug) limits the number of testable windows, particularly for women with few recorded cycles. With a mean of 13.2 cycles per woman, longer patterns such as Dilug-in-Dilug are likely underpowered.

6. Conclusion

This study provides the first rigorous statistical test of whether halachic vessel patterns are overrepresented in empirical menstrual cycle data. The five patterns analysed form a polynomial hierarchy whose structure generates mathematically necessary dependencies, confirmed by pairwise chi-square tests. Permutation and multinomial null models — run for 50,000 iterations each and in close agreement throughout — establish that Haflaga ($p < .001$) and Dilug ($p = .043$) are significantly overrepresented, while a joint multivariate test confirms collective overrepresentation of all five patterns ($p < .01$). The Haflaga result is particularly compelling: only 2 of 50,000 permutation simulations reached the observed count. A focused joint test on the three mathematically dependent patterns (Haflaga, Week-Dilug, Dilug-in-Dilug) is similarly significant ($p < .01$). These findings provide empirical statistical foundations for two of the most frequently applied halachic vessel rules, and introduce a joint distance-based test for dependent count vectors as a methodological contribution with potential applications beyond the halachic context.

AI Assistance Disclosure

Portions of the written text in this article were drafted and edited with the assistance of Claude (Anthropic, 2024), a large language model AI assistant. All scientific content, mathematical derivations, statistical analyses, and intellectual contributions are solely the work of the author. The AI tool was used exclusively for language editing, structural organisation of the manuscript, and generation of $\text{\LaTeX}$ and Python code under the author's direct supervision and verification. The author takes full responsibility for the accuracy and integrity of all content presented in this article.

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